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OPINION PAPER

Plants, soil, and coexistence: an alternative approach to modelling plant-soil feedback

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Abstract

Background and Aims Plant-soil feedback (PSF) affects plant coexistence and diversity. Although the current paradigm provides a direct experimental method to quantify PSF and explains coexistence with varying levels of success, the emphasis on pairwise interactions hinders linking PSF to community patterns.

Methods We propose a new framework emphasizing the role of individual plants in soil dynamics. Our framework suggests a common-garden experiment to quantify PSF which is species-specific rather than focused on species pairs, which require fewer treatment groups. We illustrate how data from such an experiment can be used to quantify PSF using simulated data of plant-soil dynamics.

Results We show that the current approach is susceptible to mispredicting coexistence due to incomplete characterization of the timescale of soil conditioning. Our proposed framework remedies this issue and expands the realm of possible coexistence outcomes dependent on the soil's history. Our proposed experiments are designed to confront the implicit assumptions in current PSF research and reveal gaps in our understanding of how plants condition the soil.

Conclusion The shift in focus from pairwise PSF measures to species-based soil conditioning offers a better path to linking PSF to plant coexistence and community structure, and to forecasting the effects of PSF under changing environments.

Keywords Plant-soil feedback · Trait patterns · Context-dependence · Coexistence · Community assembly · Plant diversity

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Introduction

Plant interactions with soil organisms and abiotic properties fundamentally shape community composition. Habitat filtering, whereby soil conditions determine species survival and reproduction, is arguably the best understood of these processes (Keddy 1992; Keddy & Laughlin 2021). However, plants can also modify soil conditions, directly via root exudates (Hinsinger 2001) or indirectly via interactions with microbes (Ehrenfeld, et al. 2005; Wardle, et al. 2004; Bever, et al. 2010), constituting a bidirectional interaction known as plant-soil feedback (PSF). PSF plays a significant role in mediating plant coexistence and maintaining plant diversity (van der Putten, et al. 2013; Averill, et al. 2019). Quantifying this role requires a theoretical framework connecting species' ability to condition the soil to community-level abundance patterns.

The current PSF paradigm builds upon a foundational model describing plant dynamics via soil microbiome composition (Bever, et al. 1997). This model's success stems from its straightforward experimental setup needed to parameterize it and predict coexistence outcomes (Bever, et al. 1997; Crawford, et al. 2019) (Fig. 1a). However, the focus on the uniqueness of each species' microbiome has led to the conceptualization of soil as a metric-free space, i.e., there is no notion of similarity among species' soil microbiomes. This is a limitation since plants may share general adaptations to soil conditions, e.g., acidophiles, associations with specific microbial taxa. PSF measurements have diversified over the last three decades to accommodate practical field considerations and confounding aboveground processes (Bennett, et al. 2017; De Long et al. 2023; Eppinga, et al. 2018). However, all approaches that share our goal of improving PSF-based coexistence predictions build upon the model that forms the basis of the current PSF paradigm (Bever, et al. 1997).

Since species are assumed to have distinct microbiomes, PSF interactions are defined on a pairwise basis, and a single species' soil preference cannot be characterized independently. Indeed, the central measure is the interaction coefficient (I_s in Fig. 1a), a pairwise measure comparing plant performance in conspecifics versus heterospecifics conditioned soil. Thus, the number of experiments required to quantify PSF scales quadratically with species count, making PSF quantification in multi-species communities impractical.

Furthermore, focusing on unique microbiome associations overemphasizes the frequency-dependent components of PSF at the expense of density-independent factors, such as microbial effect on plant's intrinsic growth rates (Beckman, et al. 2022; Yan, et al. 2022; Kandlikar, et al. 2019; Kandlikar, et al. 2021). As we will show, this can lead to inaccurate forecasts of long-term community coexistence outcome.

Finally, the lack of a metric for soil space leaves no framework for incorporating the timescales of plant-soil feedbacks, which could significantly impact coexistence outcomes (Ke & Levine 2021; Ke, et al. 2021; Chung 2023; Lepinay, et al. 2018). In the short term, varying soil conditioning rates can generate priority effects, while geochemical processes may continue to affect soil after mortality (Senthilnathan & D'Andrea 2023). Over longer timescales, PSF is highly context-dependent (Dudenhöffer, et al. 2022; Pugnaire, et al. 2019; De Long et al. 2019; Fry, et al. 2018; J Gundale & Kardol 2021; Crawford & Hawkes 2020). For example, greenhouse experiments indicate that PSF may promote coexistence under drier conditions but destabilize it under wetter ones (Dudenhöffer, et al. 2022). The current paradigm cannot address this context dependence directly, treating different contexts as independent scenarios through repeated pairwise experiments.

To address these challenges, we propose an alternative framework that conceptualizes soil as a continuous metric space (Fig. 1b), parameterized by a slight modification to current experiments (Fig. 1c). In the section "Towards a new conceptualization of the soil", we outline the transition from the pairwise interaction coefficient to evaluating each species' intrinsic growth rate as a function of the soil condition. In the section "Coexistence predictions", we show three scenarios where the current paradigm mispredicts coexistence while our proposed framework succeeds. In the section "Proposed experimental design", we describe the new experimental approach. Finally, we discuss the benefits of our framework and how future directions can address its limitations.

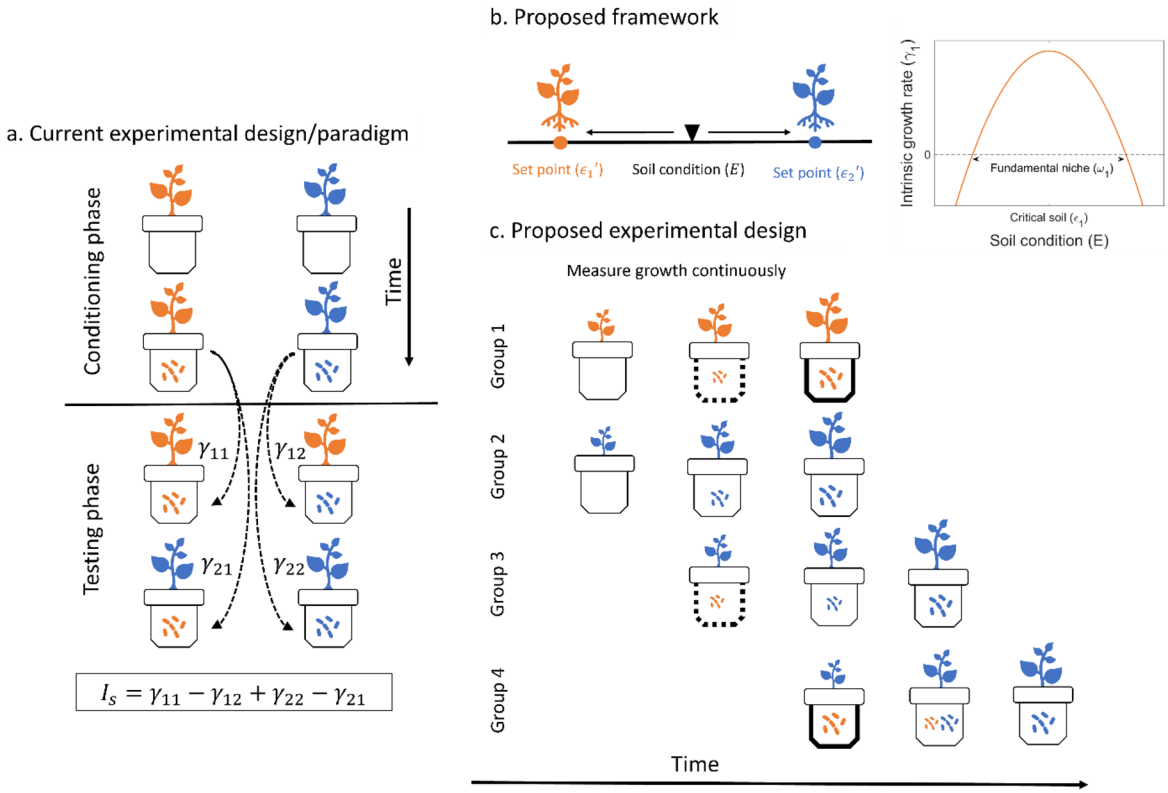


Fig. 1 Comparison of the two PSF approaches. (**a**) Two phases of the current experimental design. In the conditioning phase, plants are grown in pots with no species-specific soil inoculum to prepare conditioned soil. The conditioned soil is then used in the testing phase to measure intrinsic growth rate in conspecific and heterospecific soil conditions. γ_{ij} represents the performance measurement of plant species i in soil conditioned by plant species j. PSF is quantified using the interaction coefficient I_s. (**b**) Each species forces the soil to its conditioning setpoint (ϵ'). The growth rate of the plant depends on the soil condition (denoted by the black wedge). PSF is quantified by the mathematical function relating intrinsic growth rate and the soil condition. The critical soil condition ϵ denotes soil condition at which the growth rate is maximum/minimum. (**c**) There are still two phases in the proposed experimental design which is split into four treatment groups. In the first two groups, plant biomass is continuously measured in pots with no species-specific soil inoculum. Soil from the first treatment group with species 1 growing is collected at an intermediate point during the experiment and at the end to use as inoculate for the latter two treatment groups (shown as dotted outline and thick outline, respectively). In groups 3 and 4, species 2 biomass is continuously measured in pots with the soil inoculum which was collected

Towards a new conceptualization of the soil

Current PSF research is strongly biased towards plant–microbe interactions to the exclusion of other soil features (van der Putten, et al. 2016; De Long et al. 2023). The soil microbiome is given special treatment because it plays an outsized role in modulating plant fitness and mediating resource acquisition. Moreover, selective interactions between plants and microbes create a nonrandom association between plants and the soil microbiome (Ranelli, et al. 2015; Kivlin, et al. 2022; Klironomos 2002; Chung, et al. 2019a, b). Therefore, most PSF models assume that each plant promotes its unique microbiome, and predict coexistence outcomes using the interaction coefficient (Fig. 1a, but see (Mack, et al. 2019; Eppinga, et al. 2018; Revilla, et al. 2012)). Coexistence requires a negative interaction coefficient, meaning each plant's soil impacts conspecifics more negatively (or less positively) than heterospecifics. This is currently termed negative pairwise PSF, reflecting the negative frequency dependence that stabilizes two-species coexistence. Under this paradigm, two species cannot coexist, and priority effect

emerges under positive pairwise PSF, such as when plants grow better in soil conditioned by conspecific individuals.

The assumption of a unique microbiome for each species excludes the possibility that certain species may share similar microbial preferences and ignores its implications for community assembly. Thus, the current PSF paradigm cannot leverage the empirical literature quantifying soil community dissimilarities (Brant, et al. 2006; Burns, et al. 2015; Delavaux, et al. 2023; Malacrinò & Bennett 2024; Labouyrie, et al. 2023) to predict coexistence in the field. For instance, consider three plant species: A and B have similar microbiomes, while C's microbiome differs. If an individual grows in soil previously conditioned by species A, the rate of microbiome change will vary by species. Species B will induce minimal changes, while species C will lead to greater turnover. Furthermore, the framework's focus on the relative frequencies of species and their microbiomes means any shift in microbiome composition requires one species' microbiome to decrease while another increases. For example, if a microbe unique to species A decreases, one associated with species B must increase, regardless of conditioning rates or total microbial abundance. Addressing these issues is essential to understand species coexistence due to plant-soil feedback.

We address these issues by modeling soil as a continuous variable such as pH. Each species conditions soil to a specific pH, for example, *Sphagnum* lowers pH (van Breemen 1995). In addition, we assume there is a particular pH in which the species grows best or worst. Mathematically, each soil condition (E) is a point in a K-dimensional metric space ($K \leq N$, where N is the number of plant species), and distances represent differences between conditions. In the simplest case ($K = 1$), soil condition can be represented by a single property such as pH, or the first principal component of several soil properties, which has been found to explain a third to three-quarters of the total variation (Brant, et al. 2006; Bodenhausen, et al. 2023; Dickey, et al. 2020; Mathew, et al. 2012; Both, et al. 2018; Li, et al. 2024). This setup accommodates similarities in how species condition the soil and how they respond to soil conditions, which may reflect phylogenetic conservatism or shared traits. Species i conditions the soil to its set point ϵ_i' with strength η_i (Fig. 1b). Species respond to soil based on their intrinsic growth rate $r(E)$. For simplicity, we assume $r(E)$ is quadratic (see Box 1 and Table 1). Under this assumption, each species has a critical soil condition ϵ_i (where growth is maximized or minimized), a niche width ω_i, and a critical growth rate β_i (Fig. 1b). More complex response functions are possible but require additional parameters.

In the following sections, we show how this framework provides new insight into plant coexistence and outline an experiment to estimate these parameters.

Mathematical framework

We assume logistic growth of the plant populations N_i of species i, except their intrinsic growth rate r_i depend on the current state E of the soil. The soil in turn changes due to conditioning by plants towards the species' setpoint $\epsilon_i\prime$. Outside the plant's influence, the soil will tend towards an externally driven state ϵ_0. Our equations are thus:

$$\frac{dN_i}{dt} = N_i\left(r_i(E) - \sum\nolimits_j \alpha_{ij} N_j\right)$$

$$\frac{dE}{dt} = \eta_0\left(\epsilon_0 - E\right) + \sum\nolimits_i \eta_i N_i\left(\epsilon_i' - E\right)$$

with initial conditions

$$N_i(0) = N_{i0}$$
$$E(0) = E_0$$

where α represents the strength of negative density-dependence resulting from soil-independent factors, η_i represents the conditioning strength of the plant, and η_0 is the conditioning strength of the external drivers of the soil.

In its most general form, this model can describe a multidimensional soil state, such as the local concentrations of various nutrients and the local abundances of various microbial species. In that case, the state variables E would be a vector, and the various parameters would be vectors or matrices. Here we will stick to the simplest case of a one-dimensional soil state, for example, the soil pH.

The growth rate function $r(E)$ can assume any shape that describes plant performance in different soil conditions. We will use a quadratic r function:

Table 1 List of assumptions

#	Assumption	Is it part of the current paradigm?	How is it reflected in the proposed experiment?	How to adjust the approach if the assumption is violated?
1	Plants condition the soil	✓	Growth rate varies over time	Plants do not condition the soil. This cannot be demonstrated in the current paradigm
2	Soil conditioning is independent of height/biomass changes	✓	Height/biomass changes linearly in response to the difference between the current soil condition and conditioning setpoint	Generalize soil conditioning using a nonlinear function based on ontogeny
3	Height/biomass gain are proxies for the intrinsic growth rate of the population	✓	Variation in height/biomass gain reflects changes in intrinsic growth rates	Use alternative proxies such as seed output
4	Under a fixed soil condition, plants gain height/biomass at a constant rate	✓	Height/biomass time series has at most one inflection point, and does not saturate	Implement PSF-free treatment (e.g. sterile soil) as a point of comparison for height/biomass gain. Use field data to improve the growth model
5	Conditioning occurs to completion within experimental timescale	✓	Growth rate saturates by the end of the experiment	Run experiment for longer duration
6	Soil condition space is one-dimensional	✗	Species can only condition the soil in the same or opposite direction to that of other species	If single species fit the regression well but two species regressions do not, then soil condition space might be higher-dimensional requiring a different set of regressions (see Appendix A1)
7	(a) Soil condition decays exponentially to setpoint (b) Conditioning setpoint = critical soil condition (c) Growth curve is unimodal	✗	Proposed experiment cannot directly test those assumptions	Use additional treatment groups to test and refine those assumptions

$$r_i(E) = \beta_i\left(1 - \frac{\left(E - \epsilon_i\right)^2}{\omega_i^2}\right)$$

This assumes that each plant has one optimum soil pH (ϵ_i)—the plant's critical soil condition—at which the population grows at its maximum rate β_i (the plant's critical growth rate), and the growth rate at other soil conditions decreases proportionally to the deviation from this optimum (Fig. 1b). When the soil pH is significantly different from the optimum, mortality will exceed recruitment, leading to a negative intrinsic growth rate. The range of soil conditions in which the growth rate is positive is controlled by the niche width parameter (ω).

This scenario represents positive soil conditioning: plants improve the soil towards their preferred condition. In modeling scenarios with negative conditioning, the soil state E could be conceptualized as pathogen concentration, for example. In this case, the r function would have a minimum at the plant's critical soil condition ϵ (β would be negative).

This formalism introduces five PSF-related parameters per plant species: its conditioning setpoint (ϵ_i'), its critical soil condition (ϵ), its critical growth rate (β), its niche width (ω), and its conditioning strength (η).

Coexistence predictions

The proposed framework gives a more complete picture of coexistence outcomes than the current paradigm. We first review the coexistence condition in the current paradigm and the definition of PSF used there. We refer to it as pairwise PSF to distinguish it from our definition ascribing the sign based on whether conditioning by the plant increases (positive PSF) or decreases (negative PSF) its intrinsic growth rate. The pairwise PSF, quantified by the interaction coefficient I_s, is defined as the difference between a species' growth rate in soil conditioned by conspecifics and its growth rate in soil conditioned by heterospecifics. This measure is equivalent to conspecific frequency dependence (Vellend 2010). There are only two possible outcomes of two species interactions in the current paradigm: (i) both species coexist if pairwise PSF is negative, (ii) only one species survives if pairwise PSF is positive. Which species survives depends on the initial abundances and conditioning strengths. There is no dependence on the soil condition directly since soil condition is instantaneously determined by species frequencies (Bever, et al. 1997).

One possible outcome the current paradigm misses is that coexistence and priority effects can be alternative stable states (Senthilnathan & D'Andrea 2024). This depends not only on the sign of PSF but also on species' critical soil conditions and conditioning setpoints. We illustrate this using a graphical method for inferring coexistence outcomes in our framework (Fig. 2; analytical conditions are available for special cases, see (Senthilnathan & D'Andrea 2023)). The examples shown in Fig. 2 also demonstrate the potential for the current paradigm to mispredict coexistence.

(i) Two species coexist despite predictions of no coexistence by the current paradigm (Fig. 2a). In a community where plants condition the soil favorably ($\beta = 1$) and the two species have critical soil conditions $\epsilon_1 = -1$, $\epsilon_2 = 0$, respectively, the interaction coefficient ($I_s = (\gamma_{11} - \gamma_{12}) + (\gamma_{22} - \gamma_{21}) = 2$) is positive. Therefore, the current paradigm predicts that the species cannot coexist. However, our framework shows that the soil condition will fall between the two species as they simultaneously condition the soil, and they will coexist since both have positive intrinsic growth rates within this range of soil conditions. The current paradigm misses coexistence here because soil conditioning can have runaway effects when pairwise PSF is positive: as soil condition is equated with the frequency of one species in two-species communities, any bias towards one species gets amplified until only one species remains. By contrast, our framework treats soil as an independent variable and focuses on species density instead of frequency. This decoupling prevents runaway amplification, allowing both species to coexist.

(ii) Two species cannot coexist despite predictions of coexistence by the current paradigm (Fig. 2b). When critical growth rate is positive for species 1 (blue species) but negative for species 2 ($\beta_1 = 1, \beta_2 = -1.2$) and the two species have critical soil conditions $\epsilon_1 = -1, \epsilon_2 = 0.2$, the interaction coefficient ($I_s = -0.288$) is negative, and thus the current paradigm predicts that the species will coexist. However, for soil conditions between the critical soil condition of the two species, species 2 will always go extinct since it has a negative growth rate. This occurs

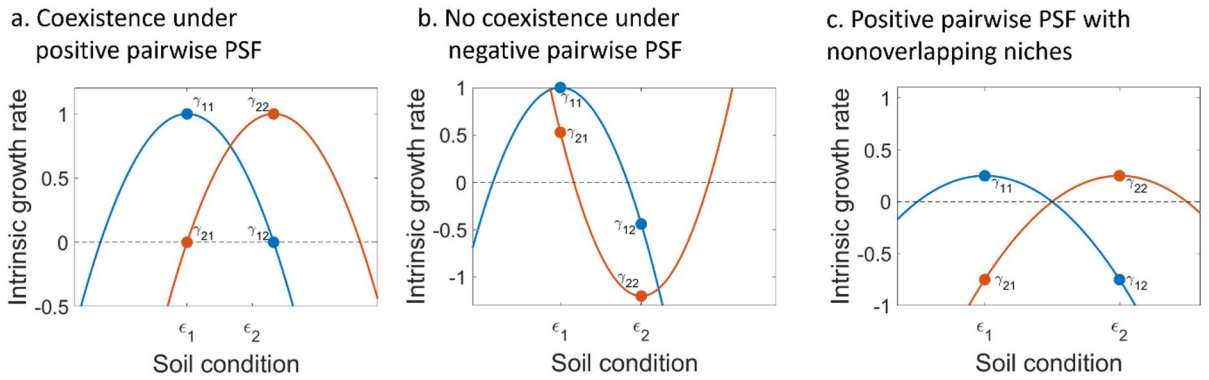


Fig. 2 Coexistence outcomes. Each panel shows the growth curves r(E) of species 1 (blue) and species 2 (orange). (**a**) Example where the current paradigm predicts exclusion, but our proposed framework shows that species coexist. (**b**) Example where the current paradigm predicts coexistence, but our proposed framework shows extinction of species 2. Here, the growth curve for species 2 has a local minimum between two peaks lying outside the displayed soil range. This can arise if plant growth accumulates specialized pathogens that drive the soil toward an intermediate state. (**c**) Example with the same interaction coefficient as (**a**) but where species' fundamental niches do not overlap and thus, they cannot coexist. Note that because the growth functions allow negative growth at some soil conditions, experiments must include replicates where plants die, since individual growth is being used as a proxy for population growth

because the current paradigm excludes the possibility of soil conditions where population growth is negative.

(iii) Two species pairs might have the same interaction coefficient yet exhibit different coexistence outcomes in our framework. For example, both species pairs in Fig. 2a and 2c have $I_s = 2$. However, the first pair of species have significant fundamental niche overlap and will coexist while the second pair ($\epsilon_1 = -1, \epsilon_2 = 1$) have non-overlapping fundamental niches. This is again due to the soil's representation as the frequency of species in the community. The soil niche axis in the proposed framework is essential to reveal the difference between such communities, which are indistinguishable from the pairwise PSF standpoint.

Proposed experimental design

While our approach to modeling PSF is new, the same two-phase conditioning experiments can be used to parameterize it, with some slight modifications (Fig. 1c). In the current paradigm, plant performance is measured only in soil that is considered to be fully conditioned. By contrast, the proposed framework requires measuring plant performance more frequently during the conditioning process.

In brief, for a simple two-species case with a one-dimensional soil axis ($K = 1$), no external drivers ($\eta = 0$ making ϵ_0 irrelevant), and a critical soil condition matching the conditioning setpoint ($\epsilon_i' = \epsilon_i$), the approach consists of the following steps. Scale the soil condition axis E such that the origin is the unconditioned soil (e.g., potting mix without any species-specific soil inoculum) and the conditioning setpoint of the first species is set as $\epsilon_1' = -1$. Estimate the three remaining parameters for species 1 by measuring plant height or other fitness proxies against time to fit the function $H_1(\beta_1, \eta_1, \omega_1, t) = \int_0^t h_1(E)\mathrm{d}t$, where $H_1(t)$ is the plant's height at time t, and h_1 is the soil-dependent height gain function. Similar to the current paradigm, we assume that this height gain function is an estimate of the intrinsic growth rate r_1. We also assume that soil conditioning is independent of plant height and depends only on the difference between the current soil condition and the conditioning setpoint. However, one must account for survival, as high mortality can lower the average height and result in negative r, even if surviving individuals grow well. We specifically chose fitness proxies which do not require destructive sampling to minimize the number of replicates required. We note that different proxies may be more appropriate for different species (Chung, et al. 2019a, b; Dudenhöffer, et al. 2018). We make a few other assumptions, listed in Table 1. Estimating

the parameters of species 2 requires three additional time series, obtained by growing it in three different soil conditions (see Fig. 1c, Appendix A1). We provide a worked-out example in Box 2.

Scenarios where $\epsilon_i' \neq \epsilon_i$ or $K > 1$ require more experiments to fit the new parameters. For example, for two species, assuming $\epsilon_i' = \epsilon_i$ requires fitting five parameters, whereas relaxing this assumption increases this to seven free parameters and thus more experimental treatments (see Appendix A2 for some directions).

Proof-of-concept of the experimental design

We use data simulated from the model in Box 1 with added noise (see Appendix A1 for details). Because the soil condition axis is only a reference frame for comparing plant responses, we set the initial soil condition of the two uninoculated treatments (blank pots in groups 1 and 2; Fig. 1c) to $E = 0$. We also choose the axis scale so that the conditioning setpoint (ϵ_1') of species 1 is $E = -1$.

We establish four treatment groups that differ in initial soil (Fig. 1c). In group 1 (orange time series in Fig. 3), species 1 grows in uninoculated potting mix; as it conditions the soil, E moves from 0 toward -1. We track height or some other fitness proxy through time and fit the resulting time series to the intrinsic growth rate curve. When averaging replicates, we account for survival, as high mortality reduces mean height and can yield negative r even if survivors grow well. Negative growth can also reflect individual height loss (wilting, drying, etc.), not just demographic change.

Curvature in the time series of average height indicates response to conditioning; no curvature suggests no conditioning or no response. Full conditioning corresponds to a stabilized growth rate (an approximately linear time series), although reaching full conditioning is not required for fitting; the duration of the experiment must be long enough to observe changes in plant growth rate reflecting changes in the soil due to plant conditioning. Table 1 lists assumptions and how the experiment can test them. Assuming that the intrinsic growth rate is a quadratic function of soil condition, group 1 yields species 1's niche width and hence a full fit of its intrinsic growth curve (Appendix 1 links regression coefficients to PSF parameters).

To estimate the remaining parameters for species 2 (its conditioning setpoint and niche width), we use groups 2–4, where species 2 grows under three initial soil conditions (blue circles in Fig. 3). Group 2 (uninoculated soil) places species 2's conditioning setpoint relative to $E = 0$ (it may lie on either side). Group 4 (soil fully conditioned by species 1) locates the species' setpoints relative to each other but still

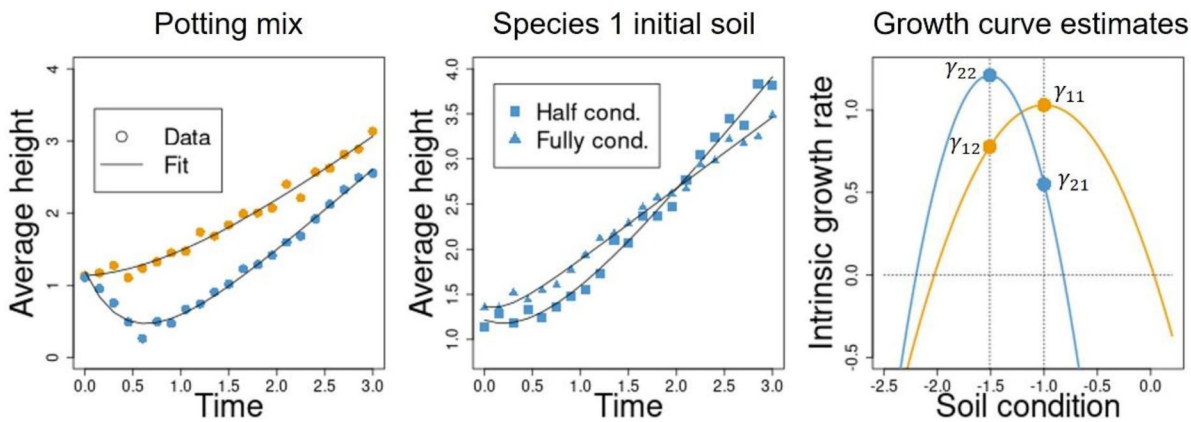


Fig. 3 Worked-out example using simulated data. The time series in the left panel are from experimental groups with species 1 (orange) and species 2 (blue) grown in potting mix. The time series in the middle panel are from experimental groups with species 2 growing in soils conditioned by species 1 to varying levels. The data is fitted by nonlinear regression (the nonlinear function and its derivation are in Appendix 1). The soil niche width of species 1 is estimated from the nonlinear regression of the first time series (in orange). Nonlinear regression of the other three time series can be used to estimate the conditioning setpoint and niche width of species 2. Finally, the estimated parameters can be used to obtain the intrinsic growth curves for species 1 (orange) and species 2 (blue)

cannot determine whether species 2's setpoint lies left or right of species 1 ($E = -1$). Group 3 (soil half conditioned by species 1) resolves both ambiguities and pins down the relative positions of both conditioning setpoints. Testing our proposed experiment with real data is an important next step.

Other benefits of the new concept

One problem with the interaction coefficient is that it equates soil microbiome differences entirely with niche differences, leaving out fitness differences (Beckman, et al. 2022; Yan, et al. 2022). According to modern coexistence theory, two species can stably coexist when stabilizing effects from niche differences are stronger than the effect of fitness differences (Chesson 2000). However, current experiments quantifying interaction coefficient (I_s in Fig. 1a) cannot measure fitness differences without an empirical definition of "unconditioned" soil, which has proved challenging (Kandlikar, et al. 2019; Ke & Wan 2020). By modeling soil condition as a metric space, our framework can predict coexistence without directly estimating niche and fitness differences. Instead of assuming "unconditioned" soil is microbe-free, we define a system-specific reference soil, such as post-disturbance soil in a newly invaded landscape or the residual soil after plant-induced changes dissipate under geochemical forcing. Coexistence predictions then depend only on soil-dependent intrinsic growth curves, $r(E)$, estimated by our proposed experiment (Fig. 1c).

Most of our assumptions are already part of the current paradigm (Table 1), but our design also tests them directly. For example, in the current paradigm, we cannot reject the hypothesis that plants condition the soil, even when the interaction coefficient is 0, while our proposed experiment will detect it from the height time series. While current studies often assume microbial dynamics are much faster than plant dynamics, our approach explicitly represents the microbial community via the soil condition E. In plant communities dominated by associations with arbuscular mycorrhizal fungi (AM), E could represent the degree of root colonization rates, sufficing for a one-dimensional soil axis. More complex microbiomes may require higher-dimensional representations of the soil condition (see Appendix A1), providing a clear direction for future mathematical framework and experimental designs.

Importantly, our framework shows that coexistence and priority effects can be alternative stable states where the current paradigm allows only one or the other (see Appendix A3) (Senthilnathan & D'Andrea 2024). Under positive PSF, two species with very different critical soil condition and narrow niche overlap will coexist if the initial soil condition falls within that overlapping region. Alternatively, if the initial soil condition is sufficiently biased towards one species, then that species excludes the other, constituting a priority effect. If the initial soil condition is far outside both fundamental niches, then neither species survive. Therefore, we can use PSF to explain variable community compositions based on historical soil conditions.

Another advantage of our framework is its capacity to quantify context-dependence by explicitly linking environmental variation to growth rate parameters. Environmental deviations from a historical baseline (e.g., precipitation anomalies from P_0) can shift a plant's conditioning setpoint ϵ' away from its critical soil condition ϵ (e.g., $| \epsilon' - \epsilon | \propto | P - P_0 |$), allowing quantitative interpolation and extrapolation of growth rate function across climate scenarios.

Our approach also reduces experimental burden. For an N-species community, the current paradigm would require N^2 treatment groups, whereas our framework requires $3N - 2$ (1 for the first species and 3 for every other species). For example, pairwise PSF vary across eight prairie species under three watering treatments (Dudenhöffer, et al. 2022), but because environmental effects were not tied to PSF parameters, the conclusions were necessarily qualitative. By linking precipitation to parameters such as ϵ', our framework yields quantitative predictions while requiring only $3 \times (3 \times 8 - 2) = 63$ treatment groups (3 for the watering treatments) instead of $3 \times 8^2 = 192$.

Finally, focusing on direct plant-soil interactions help decouple mechanisms that the current paradigm conflates. For example, negative pairwise PSF can result from mutualist or pathogen accumulation (Chung, et al. 2023; Abbott, et al. 2021; Bever 2002). Our approach distinguishes these mechanisms using the critical growth rate β and the mismatch between critical soil condition ϵ and conditioning setpoint ϵ'. When mutualist accumulation dominates conditioning, the critical growth rate

is positive. Without mismatch between the critical soil condition and the conditioning setpoint, pairwise PSF is positive. We expect this to be the case in forests where species typically associate with ectomycorrhizal (EM) fungi (Bennett, et al. 2017). However, mutualisms do not necessarily imply positive pairwise PSF: AM fungi can benefit neighbors while only the host bears the costs, producing negative (stabilizing) pairwise PSF.

Alternatively, when pathogen accumulation dominates conditioning and the critical soil condition matches the conditioning setpoint, both the critical growth rate and the pairwise PSF are negative, as the soil becomes saturated with host-specific pathogens. Avoiding this buildup explains why crop rotations increase yield and reduce disease. However, when pathogens are generalists, particularly when the critical soil condition does not match the condition setpoint (i.e., a plant harbors pathogens that are disproportionately detrimental to other species), pairwise PSF can be positive and destabilize the community.

As the examples above illustrate, our species-specific PSF approach enables a PSF-based plant community classification scheme (Appendix Fig. A3A) that readily scales from individuals to communities. Appendix A4 provides testable predictions in multispecies communities for the spatial distribution of species and coexistence between strong and weak competitors.

Limitations and future directions

This new framework has its own limitations. First, despite our focus on a one-dimensional axis, soil condition is multidimensional. A multidimensional extension of the model is mathematically straightforward, albeit at the cost of a proliferation of parameters (Appendix A1). This can be mitigated via dimensional reduction analysis, which can provide lower-dimensional sufficient representations of the soil condition (John, et al. 2007). Experiments can also aid in lowering soil dimensionality by revealing constraints in the trajectory of soil dynamics. Moreover, our proposed experiment will confront several assumptions commonly made in plant-soil feedback studies, some of which might expose limitations for future studies to address (Table 1). Future theoretical work should focus on identifying the appropriate nonlinear functions to model the intrinsic growth rate function and the soil conditioning function.

Second, while our species-centric approach enables a trait-based analysis of PSF in plant communities, our parameters represent integrative plant attributes, such as a plant's critical soil condition, which can be difficult to relate to measurable plant functional traits such as root properties (Bergmann, et al. 2020; Rutten & Allan 2023; Xi, et al. 2021). Bridging this gap is an important direction for PSF research, and this bridge can be built by empirically linking species traits and spatial distribution to post-dimensional-reduction local soil conditions (Kirby 2000; Guerrero-Ramírez, et al. 2021; Kattge, et al. 2020; Schimel, et al. 2007). Our conceptualization of soil condition, and the use of integrative plant attributes can be further refined as future experimental studies unveil the biological mechanisms of soil conditioning and describe the effects of soil factors on plants.

Important belowground factors to address include the effects of mixed rooting neighborhoods and the spatial extent of mycorrhizal effects on plants. A spatial version of our framework can aid empirical studies in tying these factors to coexistence. Furthermore, models following the current paradigm already aim to expand into frequency-dependent effects driven by factors other than PSF (Eppinga, et al. 2018). We should seek to integrate PSF with aboveground processes, such as herbivory and light competition, by connecting our framework to other models of plant community dynamics (De Long et al. 2023).

Conclusion

Here, we introduced a new framework to advance PSF research. Our conceptualization of soil condition as a continuous state variable offers an alternative foundation for understanding plant-soil feedback. Modelling intrinsic growth rate as a function of soil condition, which is the main shift in the mathematical framework, can be used graphically to develop insights into the effects of plant-soil feedback on coexistence (Fig. 1b, Box 1). Combined with the new experimental design for parameterizing the intrinsic growth rate and soil conditioning using our new soil condition axis, this approach facilitates scaling individual-level PSF to community patterns. The main difference between our proposed experiment and the current experiment is the replacement of a conditioning phase treatment with an additional feedback phase treatment, along with continuous measurements during the experiment (Fig. 1c). By mirroring PSF's empirical context-dependence, our PSF framework

serves as an instrumental predictor of climate change impacts on plant communities.

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Data availability All codes used to make figures and demonstrate parametrization with simulated data are archived at Zenodo (Senthilnathan 2026).

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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